

AgriSmart AI – Requirements Gathering

Project: AgriSmart AI – Intelligent Crop Recommendation System **Document Type:** Requirements Gathering (DEPI Project Documentation) **Phase:** Requirements – precedes System Analysis & Design

1. System Overview

AgriSmart AI is an intelligent crop recommendation system. The user enters soil and environmental parameters, and a Machine Learning (ML) model recommends the most suitable crop for those conditions. The system also reports how confident the model is, lists alternative candidate crops, and explains which factors influenced the recommendation.

This document defines what the system must do and the qualities it must have. It does not describe how the system will be built; design and implementation decisions belong to the later System Analysis & Design phase.

1.1 Inputs and Outputs

Category	Item	Description
Input	Nitrogen (N)	Soil nitrogen content
Input	Phosphorus (P)	Soil phosphorus content
Input	Potassium (K)	Soil potassium content
Input	Temperature	Environmental temperature
Input	Humidity	Environmental humidity
Input	pH	Soil acidity or alkalinity
Input	Rainfall	Environmental rainfall
Output	Recommended crop	The single most suitable crop predicted by the model
Output	Prediction probability	The model's confidence in the recommended crop
Output	Top candidate crops	A ranked list of the most likely crops
Output	Prediction explanation	The factors that influenced the recommendation

2. Stakeholder Analysis

Stakeholder	Role	Needs	Expectations
End User / Farmer	Provides input parameters and uses the recommendation	A simple way to obtain a crop recommendation from soil and environmental data; understandable results	Clear input fields, a fast and understandable result, and an explanation of why the crop was recommended
Project Team	Plans, develops, tests, and documents the system	Clear, testable requirements and a defined scope	Requirements that are unambiguous, feasible within the 8-week schedule, and verifiable through testing
Project Supervisor / Lecturer	Guides and evaluates the project	Evidence of a structured, well-documented process	Complete documentation, consistent scope, and a working, explainable system
DEPI	Defines the project documentation and submission requirements	Documentation that follows program requirements	Professional, complete, and consistent project deliverables

3. User Stories

ID	User Story
US-01	As a user, I want to enter soil parameters (N, P, K, pH) so that the system can recommend a suitable crop.
US-02	As a user, I want to enter environmental parameters (temperature, humidity, rainfall) so that the recommendation reflects local conditions.

ID	User Story
US-03	As a user, I want to be told when my input is missing or invalid so that I can correct it before receiving a recommendation.
US-04	As a user, I want to see the recommended crop so that I know which crop suits my conditions.
US-05	As a user, I want to see the prediction probability so that I can understand how confident the model is.
US-06	As a user, I want to see the top candidate crops so that I can consider alternatives to the main recommendation.
US-07	As a user, I want to see an explanation of the prediction so that I understand the factors affecting the recommendation.
US-08	As a user, I want to reset the inputs so that I can quickly try a new set of conditions.

4. Use Cases

The only actor in the system is the **User**. A use case diagram is not included here and will be produced in the System Analysis & Design phase.

ID	Use Case	Actor	Description	Expected Result
UC-01	Enter Parameters	User	The user enters the soil and environmental values.	Entered values are captured and shown in the input fields.
UC-02	Validate Input	User	The system checks that all required values are present and acceptable.	Invalid or missing values are identified before prediction; valid input proceeds.
UC-03	Request Crop Recommendation	User	The user asks the system to generate a recommendation from the entered values.	The system processes the input and produces a prediction.

ID	Use Case	Actor	Description	Expected Result
UC-04	View Recommended Crop	User	The user views the crop predicted by the model.	One recommended crop is displayed.
UC-05	View Prediction Probability	User	The user views the model's confidence for the recommended crop.	A probability value is displayed with the result.
UC-06	View Top Candidate Crops	User	The user views the most likely crops in ranked order.	A ranked list of candidate crops with their probabilities is displayed.
UC-07	View Prediction Explanation	User	The user views the factors that influenced the prediction.	An explanation of the contribution of the input parameters is displayed.
UC-08	Reset Input	User	The user clears the entered values and any displayed results.	Inputs and results are cleared, and the system is ready for new input.

5. Functional Requirements

ID	Requirement	Related Use Case
FR-01	The system shall allow the user to enter the required soil parameters: nitrogen, phosphorus, potassium, and pH.	UC-01
FR-02	The system shall allow the user to enter the environmental parameters: temperature, humidity, and rainfall.	UC-01
FR-03	The system shall validate user inputs before a prediction is made.	UC-02
FR-04	The system shall process the validated input using the trained ML model.	UC-03
FR-05	The system shall return a recommended crop.	UC-04
FR-06	The system shall display the prediction probability of the recommended crop.	UC-05

ID	Requirement	Related Use Case
FR-07	The system shall display the top candidate crops.	UC-06
FR-08	The system shall provide an explanation of the prediction.	UC-07
FR-09	The system shall allow the user to reset the inputs.	UC-08
FR-10	The system shall display a clear message identifying which inputs are missing or invalid when validation fails.	UC-02
FR-11	The system shall not produce a prediction when validation fails.	UC-02, UC-03

FR-10 and FR-11 are added because they are necessary to make input validation (FR-03) observable and testable.

6. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	The system shall return the result of a single prediction quickly enough for interactive use. The target response time will be defined and measured during testing.
NFR-02	Usability	The system shall provide a simple interface that a non-technical user can operate without training.
NFR-03	Usability	The system shall provide clear labels for every input field, including the parameter name and its expected unit.
NFR-04	Usability	The system shall present results in an easy-to-understand form, with the recommended crop shown most prominently.
NFR-05	Reliability	The system shall validate all inputs before processing.
NFR-06	Reliability	The system shall handle errors without crashing and shall display a meaningful message to the user.

ID	Category	Requirement
NFR-07	Maintainability	The system code shall be modular and organized so that components can be understood, tested, and modified independently.
NFR-08	Security	The system shall validate user inputs to prevent invalid or unsafe values from being processed.
NFR-09	Security	The system shall not expose sensitive system information, such as internal file paths or technical error details, to the user.
NFR-10	Availability	The application shall be accessible to users when deployed.

7. Functional vs. Non-Functional Summary

Requirement Type	Purpose	Examples
Functional	Define what the system does	Parameter input, input validation, crop prediction, probability display, top candidate crops, prediction explanation, reset, result display
Non-functional	Define how well the system performs its functions	Performance, usability, reliability, security, maintainability, availability

8. Requirements Prioritization

Requirements are classified as High, Medium, or Low. High priority items are essential for a working system; Medium priority items add the interpretability and decision support that define the project's goals; Low priority items are outside the current scope.

Priority	Requirement	Related IDs	Rationale
High	Parameter input	FR-01, FR-02	The system cannot function without input
High	Input validation	FR-03, FR-10, FR-11, NFR-05, NFR-08	Prevents invalid predictions and errors

Priority	Requirement	Related IDs	Rationale
High	Crop prediction	FR-04, FR-05	Core purpose of the system
High	Result display	FR-05, NFR-04	The user must be able to see the outcome
Medium	Prediction probability	FR-06	Helps the user judge model confidence
Medium	Top candidate crops	FR-07	Provides alternatives to the main recommendation
Medium	SHAP explanation	FR-08	Supports the project objective of explainable predictions
Medium	Reset inputs	FR-09	Improves convenience; not required for a prediction
Low	Future enhancements	None	Outside the current scope; not planned in this phase

Non-functional requirements not listed above (for example NFR-01 to NFR-04, NFR-06, NFR-07, NFR-09, and NFR-10) will be addressed throughout development and verified during testing.

9. Requirements Acceptance Criteria

Criteria are written so that each can be checked by a test case.

Requirement	Acceptance Criteria
Parameter input (FR-01, FR-02)	The interface provides a field for each of the seven parameters, and entered values are accepted by the system.
Input validation (FR-03, FR-10, FR-11)	Given missing, non-numeric, or out-of-range values, the system detects them before prediction, identifies the affected fields, and does not produce a recommendation.
Crop recommendation (FR-04, FR-05)	Given valid input parameters, the system returns exactly one recommended crop.

Requirement	Acceptance Criteria
Prediction probability (FR-06)	The probability for the recommended crop is displayed together with the result and is a valid value between 0 and 100%.
Top candidate crops (FR-07)	A ranked list of candidate crops is displayed, with the recommended crop appearing first.
Prediction explanation (FR-08)	For each prediction, the system displays an explanation showing how the input parameters influenced the recommendation.
Reset (FR-09)	After the user resets, all input fields and displayed results are cleared.
Performance (NFR-01)	A single prediction returns a result within the response-time target defined during testing.
Usability (NFR-02 to NFR-04)	Every input field has a clear label and unit, and a test user can obtain a recommendation without instructions.
Reliability (NFR-05, NFR-06)	Invalid input and handled error conditions produce a clear message and do not crash the application.
Security (NFR-08, NFR-09)	Invalid input is rejected, and error messages shown to the user do not reveal internal system details.
Maintainability (NFR-07)	Code is organized into separate, documented modules that can be reviewed independently.
Availability (NFR-10)	When the application is deployed, a user can open it and request a recommendation.

10. Conclusion

This document defines the expected functionality, quality attributes, and user interaction of AgriSmart AI. It identifies the stakeholders, expresses user needs as user stories and use cases, specifies functional and non-functional requirements, assigns priorities, and sets acceptance criteria that can be verified through testing. The requirements remain within the agreed project scope and avoid design and implementation decisions. They will serve as the foundation for the next phase, System Analysis & Design, in which the use case diagram, system structure, and detailed design will be developed.